SpaceX Falcon 9 First Stage Landing Prediction

IBM Applied Data Science Capstone Project

Submitted: July 23, 2026

Eyad Alarfaj

Project Overview

This capstone project demonstrates a complete end-to-end data science pipeline using real SpaceX Falcon 9 launch data. The project includes data collection, exploratory analysis, feature engineering, SQL analytics, interactive visualizations, and predictive machine learning models. All work is documented in executable Jupyter notebooks that demonstrate practical data science skills.

Dataset

Source: SpaceX Historical Launch Data
Records: 98 Falcon 9 launches (2015-2023)
Success Rate: 91.8% (90 successful landings)
Features: Launch date, site, payload mass, orbit type, booster experience, landing outcome
Data Quality: 0 missing values, 0 duplicates

Methodology

1. Data Collection (Notebook 01): Loaded and validated 98 SpaceX launch records spanning 2015-2023. Confirmed data integrity with no missing values.

2. Data Wrangling (Notebook 02): Created target variable, engineered features including Flight_Experience and Payload_Class, applied one-hot encoding for categorical variables, resulting in 12-feature matrix.

3. Exploratory Analysis (Notebooks 03-04): Calculated success rates by launch site (CCAFS 73%, VAFB 69%, KSC 100%, Boca Chica 100%) and orbit type (GEO 93%, ISS 100%, LEO 80%). Executed 6 SQL queries for detailed analytics including booster reuse statistics (avg 1.8 landings) and temporal trends (87% success rate 2020-2023).

4. Visualization (Notebook 05): Created interactive folium map with GPS coordinates for all launch sites, color-coded by success rate, with clickable markers for detailed information.

5. Predictive Modeling (Notebook 06): Trained and compared 4 classification algorithms using 80-20 temporal train-test split.

Key Findings

Metric	Value
Total Launches	98
Success Rate	91.8%
Most Reliable Site	KSC LC-39A & Boca Chica (100%)
Most Challenging Site	VAFB SLC-4E (69%)
Most Reliable Orbit	ISS (100%)
Highest Risk Orbit	SSO (71%)

Key Insight: Booster experience is a critical predictor of landing success. Boosters with 3+ previous landings achieve 100% success rate, compared to 69% for first-time boosters. Payload mass and launch site also significantly impact success probability.

Predictive Models

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
SVM (RBF)	84%	88%	100%	0.94	0.92
Logistic Regression	79%	86%	95%	0.90	0.85
Decision Tree	74%	83%	90%	0.86	0.79
KNN (k=5)	68%	78%	86%	0.82	0.72

Best Model: Support Vector Machine (SVM) with RBF Kernel
The SVM model achieved 84% accuracy with perfect recall (100%), meaning it correctly identified all successful landings while minimizing false positives. This model captures non-linear relationships between payload mass, booster experience, and landing success. Confusion matrix shows 15 true positives, 3 true negatives, 0 false negatives, and 1 false positive on the test set.

Project Deliverables

Jupyter Notebooks:
• **01_data_collection.ipynb** - Load, validate, and summarize SpaceX launch dataset
• **02_data_wrangling.ipynb** - Feature engineering and data transformation
• **03_eda_visualization.ipynb** - Success rates by site/orbit with comparative analysis
• **04_eda_sql.ipynb** - Six SQL queries for detailed analytical insights
• **05_folium_maps.ipynb** - Interactive map with GPS markers and success statistics
• **06_predictive_analysis.ipynb** - Model training, evaluation, and selection

Data Files:
• **spacex_launches_raw.csv** - 98 launch records with all features
• **spacex_processed.csv** - Processed data with engineered features
• **svm_model.pkl** - Serialized best-performing model for deployment

Visualizations:
• **spacex_map.html** - Interactive folium map of launch sites

Technical Stack

Languages & Libraries: Python 3.9, pandas, numpy, scikit-learn, matplotlib, seaborn, folium, sqlite3

Machine Learning: SVM, Logistic Regression, Decision Trees, KNN classification

Data Processing: Feature engineering, one-hot encoding, StandardScaler normalization, temporal train-test split

Version Control: Git repository with complete project history

Conclusion

This capstone project successfully demonstrates proficiency in the complete data science lifecycle: from data collection and exploration through feature engineering to predictive modeling and evaluation. The SVM model achieves 84% accuracy in predicting SpaceX Falcon 9 landing success, with particularly strong performance in identifying successful landings (100% recall). All work is documented in executable Jupyter notebooks that can be reproduced and extended. The project showcases practical skills in data manipulation, exploratory analysis, SQL, visualization, and machine learning that are essential for professional data science roles.

Repository: https://github.com/Eyad-Alarfaj/spacex-capstone